Palak Deb Patra

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EDUCATION

National Institute of Technology, Rourkela **Dec 2021 – Jun 2025**
Bachelor of Technology in **Computer Science & Engineering** | **CGPA: 8.52/10** **Rourkela, India**

Coursework: *Data Structures and Algorithms, Database Engineering, Operating Systems, Computer Networks, Probability and Statistics, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Science, Object-Oriented System Design, Distributed Systems.*

RESEARCH PUBLICATIONS

IEEE SPCOM 2026 – IISc Bengaluru | First Author **Aug 2026**
P. D. Patra, S. K. Panda, M. K. Bishwal, and P. K. Sa, "Bidirectional Convolutional-GRU Integrated GAN for Unpaired Low-Light Image Enhancement," in *2026 International Conference on Signal Processing and Communications (SPCOM)*, IEEE, 2026. doi:10.1109/SPCOM71105.2026.11622932.

PROFESSIONAL EXPERIENCE

NatWest Group | **Software Engineer** **Jul 2025 – Present**
Python, FastMCP, Azure, Microsoft Copilot Studio, Power Platform **Bangalore, India**

Project: Colleague Learning Assistant (CLA)

- Built **end-to-end AI observability** using **OpenTelemetry** and **Azure Application Insights**, which enables tracing of requests, tool calls and conversations, with failures classified into **8 categories**.
- Developed **131 automated tests** to validate agent behaviour and telemetry, along with **11 Kusto queries** to investigate failures and monitor agent activity.
- Configured **Single Sign-On (SSO)** using **PingFederate** and **Microsoft Entra ID** with a **three-step fallback mechanism** that reliably identifies employees, handles missing authentication details and prevents incorrect user identification.
- Integrated **Workday learning completion records** from the **last 12 months** as an additional data source, which allows the agent to consider previously completed courses when generating personalised learning recommendations.
- Built an **Azure Blob Storage data pipeline** to parse GitLab coding metrics from Excel workbooks, which detects changes to avoid redundant downloads and uses the last valid copy when downloads fail.
- Redesigned the **UI/UX of 3 Microsoft Teams widgets** (Learning Courses, ECDF progress and GitLab metrics) to follow NatWest branding guidelines and resolve existing rendering issues.

Project: ChLoE

- Worked on migrating deployments from **Power Platform pipelines** to **GitLab CI/CD**, while ensuring the workflows follow the bank's security guardrails and release requirements.
- Reduced **agent hallucinations** by improving Power Platform FAQ knowledge sources and testing revised responses in SIT and UAT environments, with **23 out of 26 test cases passing**.
- Fixed **duplicate agent responses** after SIT/UAT deployments and corrected the **"Start over" trigger functionality** in Microsoft Teams, which clears chat history, closes active topics and starts a conversation with a new memory context.

NatWest Group | **Software Engineering Intern** **May 2024 – Jul 2024**
Python, GraphQL, REST APIs, Snowflake **Gurgaon, India**

Project: OSPO Metrics

- Collected and consolidated **engineering activity metrics** for members of our business unit using **GitLab GraphQL and REST APIs**.
- Automated data extraction and CSV exports, thereby reducing manual processing time by **28%**.
- Loaded **16 datasets** into **Snowflake** and developed a unified data model that supports interactive dashboards in **ThoughtSpot**.

PERSONAL PROJECTS

CodeShift | Claude, Python, TypeScript, LangGraph, Ollama | GitHub **Aug 2026**

- Built a **six-agent system** using LangGraph and Ollama to convert Python projects into TypeScript, converting supporting modules before the files that use them.
- Used **mypy** and **tsc** for type checking and developed **differential tests** to compare outputs, exceptions and object states between the original and translated code.
- Evaluated the system on a **31-module application**, testing **197 functions and methods** across both languages and confirming identical behaviour for **15 modules** under the generated tests.

Memex | Claude, Python, LangGraph, ChromaDB, Ollama | GitHub **Aug 2026**

- Built a **local RAG system** that answers questions about documents with source citations, supports OCR and automatically updates its index when files change.
- Combined **dense retrieval, BM25 and cross-encoder reranking** to improve document retrieval, and indexed **3,050 files** into **42,743 chunks** in approximately **15 minutes** on my local GPU.
- Evaluated retrieval on **42 questions** across a separate **23-document corpus**, ranking a relevant result first for **41 of 42 questions** and achieving **97.6% Hit@1**, **0.988 MRR** and **100% Recall@5**.

TECHNICAL SKILLS

Languages	Python, Java, C++, TypeScript, SQL
AI & ML	PyTorch, TensorFlow, scikit-learn, OpenCV, spaCy, NLTK, NVIDIA CUDA
LLMs & Agents	LangGraph, LangChain, Hugging Face, Ollama, FastMCP, Claude, Codex
Web & APIs	Angular, Streamlit, FastAPI, REST APIs, GraphQL
Databases & Cloud	Snowflake, Oracle, MySQL, ChromaDB, AWS, Azure
Tools	Git, GitLab CI/CD, Ansible, Jupyter

ACHIEVEMENTS

- Selected for the **Amazon ML Summer School 2023**.
- Solved **800+** LeetCode problems with a peak rating of **1795**, ranking in the **top 8%** globally.
- Achieved global ranks of **277** in Weekly Contest 438 and **1839** in Weekly Contest 437 on LeetCode.

LEADERSHIP & COMMUNITY SERVICE

Coordinator | Event Management | **NITRUTSAV 2023** **Aug 2023 – Sep 2023**

- Led a team of **7 volunteers** in organising NIT Rourkela's flagship cultural festival and helped secure a **20% increase in sponsorship**.

Senior Staff Member | Social Service | **AASRA** **Jun 2022 – Jan 2024**

- Supported outreach and fundraising for the **DIYA Project**, which raised **Rs 2,11,000** through collective team efforts.